

Analyzing Redline Pcap

Summary:

RedLine is a stealer distributed as cracked games, applications, and services.

The [malware](#) steals information from web browsers, cryptocurrency wallets, and applications such as FileZilla, Discord, Steam, Telegram, and VPN clients. The binary also gathers data about the infected machine, such as the running processes, antivirus products, installed programs, the Windows product name, the processor architecture, etc. The stealer implements the following actions that extend its functionality: Download, RunPE, DownloadAndEx, OpenLink, and Cmd. The extracted information is converted to the XML format and exfiltrated to the C2 server via SOAP messages.

Analysis:

Today, we'll be analyzing a Redline pcap after its initial detonation on a host and going over what IOCs there are and what to look for with this given malware.

Upon initially opening the pcap you will be greeted with a decent amount of network traffic from multiple source IPs.

2021-08-19-redline.pcap

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter: <Ctrl>

No.	Time	Source	Destination	Protocol	Length	Info
2334	606.304596	c6:ac:47:a3:50:44	de:d7:f0:66:cf:6d	ARP	42	10.10.0.1 is at c6:ac:47:a3:50:44
2333	606.304579	de:d7:f0:66:cf:6d	c6:ac:47:a3:50:44	ARP	60	who has 10.10.0.1? Tell 10.10.0.33
2332	605.580420	185.215.113.17	10.10.0.33	ICMP	94	Destination unreachable (Host unreachable)
2331	605.400044	10.10.0.33	185.215.113.17	TCP	66	[TCP Retransmission] [TCP Port numbers reused] 53533 → 18597 [SYN] Seq=64240 Len=0 MSS=1460 WS=256 SACK_PERM=
2330	600.110152	185.215.113.17	10.10.0.33	ICMP	94	Destination unreachable (Host unreachable)
2329	597.180162	185.215.113.17	10.10.0.33	ICMP	94	Destination unreachable (Host unreachable)
2328	597.070453	10.10.0.33	185.215.113.17	TCP	66	[TCP Retransmission] [TCP Port numbers reused] 53533 → 18597 [SYN] Seq=64240 Len=0 MSS=1460 WS=256 SACK_PERM=
2327	594.872044	10.10.0.33	185.215.113.17	TCP	66	[TCP Retransmission] [TCP Port numbers reused] 53533 → 18597 [SYN] Seq=64240 Len=0 MSS=1460 WS=256 SACK_PERM=
2326	577.578576	185.215.113.17	10.10.0.33	ICMP	94	Destination unreachable (Host unreachable)
2325	577.023495	10.10.0.33	185.215.113.17	TCP	66	[TCP Retransmission] [TCP Port numbers reused] 53532 → 18597 [SYN] Seq=64240 Len=0 MSS=1460 WS=256 SACK_PERM=
2324	575.804665	c6:ac:47:a3:50:44	de:d7:f0:66:cf:6d	ARP	42	10.10.0.1 is at c6:ac:47:a3:50:44
2323	575.804646	de:d7:f0:66:cf:6d	c6:ac:47:a3:50:44	ARP	60	who has 10.10.0.1? Tell 10.10.0.33
2322	573.858321	185.215.113.17	10.10.0.33	ICMP	94	Destination unreachable (Host unreachable)
2321	571.023480	10.10.0.33	185.215.113.17	TCP	66	[TCP Retransmission] [TCP Port numbers reused] 53532 → 18597 [SYN] Seq=64240 Len=0 MSS=1460 WS=256 SACK_PERM=
2320	569.778267	185.215.113.17	10.10.0.33	TCP	66	[TCP Retransmission] [TCP Port numbers reused] 53532 → 18597 [SYN] Seq=64240 Len=0 MSS=1460 WS=256 SACK_PERM=
2319	568.815147	10.10.0.33	185.215.113.17	TCP	66	[TCP Retransmission] [TCP Port numbers reused] 53532 → 18597 [SYN] Seq=64240 Len=0 MSS=1460 WS=256 SACK_PERM=
2318	566.272204	de:d7:f0:66:cf:6d	c6:ac:47:a3:50:44	ARP	60	10.10.0.33 is at de:d7:f0:66:cf:6d
2317	566.272181	c6:ac:47:a3:50:44	de:d7:f0:66:cf:6d	ARP	42	who has 10.10.0.33? Tell 184.61.71.13
2316	552.728924	185.215.113.17	10.10.0.33	ICMP	94	Destination unreachable (Host unreachable)
2315	550.991954	10.10.0.33	185.215.113.17	TCP	66	[TCP Retransmission] [TCP Port numbers reused] 53531 → 18597 [SYN] Seq=64240 Len=0 MSS=1460 WS=256 SACK_PERM=
2314	550.804676	c6:ac:47:a3:50:44	de:d7:f0:66:cf:6d	ARP	42	10.10.0.1 is at c6:ac:47:a3:50:44
2313	550.804656	de:d7:f0:66:cf:6d	c6:ac:47:a3:50:44	ARP	60	who has 10.10.0.1? Tell 10.10.0.33
2312	546.358541	185.215.113.17	10.10.0.33	ICMP	94	Destination unreachable (Host unreachable)
2311	544.992217	10.10.0.33	185.215.113.17	TCP	66	[TCP Retransmission] [TCP Port numbers reused] 53531 → 18597 [SYN] Seq=64240 Len=0 MSS=1460 WS=256 SACK_PERM=
2310	542.358310	185.215.113.17	10.10.0.33	ICMP	94	Destination unreachable (Host unreachable)
2309	540.792255	10.10.0.33	185.215.113.17	TCP	66	[TCP Retransmission] [TCP Port numbers reused] 53531 → 18597 [SYN] Seq=64240 Len=0 MSS=1460 WS=256 SACK_PERM=
2308	525.408795	185.215.113.17	10.10.0.33	ICMP	94	Destination unreachable (Host unreachable)
2307	525.304718	c6:ac:47:a3:50:44	de:d7:f0:66:cf:6d	ARP	42	10.10.0.1 is at c6:ac:47:a3:50:44
2306	525.304650	de:d7:f0:66:cf:6d	c6:ac:47:a3:50:44	ARP	60	who has 10.10.0.1? Tell 10.10.0.33
2305	524.945231	10.10.0.33	185.215.113.17	TCP	66	[TCP Retransmission] [TCP Port numbers reused] 53530 → 18597 [SYN] Seq=64240 Len=0 MSS=1460 WS=256 SACK_PERM=
2304	521.978567	185.215.113.17	10.10.0.33	ICMP	94	Destination unreachable (Host unreachable)
2303	520.440489	de:d7:f0:66:cf:6d	c6:ac:47:a3:50:44	ARP	60	10.10.0.33 is at de:d7:f0:66:cf:6d
2302	520.440309	c6:ac:47:a3:50:44	de:d7:f0:66:cf:6d	ARP	42	who has 10.10.0.33? Tell 154.61.71.13
2301	518.958593	185.215.113.17	10.10.0.33	ICMP	94	Destination unreachable (Host unreachable)
2300	518.929813	10.10.0.33	185.215.113.17	TCP	66	[TCP Retransmission] [TCP Port numbers reused] 53530 → 18597 [SYN] Seq=64240 Len=0 MSS=1460 WS=256 SACK_PERM=
2299	515.317132	10.10.0.33	185.215.113.17	TCP	66	[TCP Retransmission] [TCP Port numbers reused] 53530 → 18597 [SYN] Seq=64240 Len=0 MSS=1460 WS=256 SACK_PERM=

Frame 6: 90 bytes on wire (720 bits), 90 bytes captured (720 bits) on ethnet II, Src: de:d7:f0:66:cf:6d (de:d7:f0:66:cf:6d), Dst: c6:ac:47:a3:50:44 (c6:ac:47:a3:50:44)

Ethernet II, Src: de:d7:f0:66:cf:6d (de:d7:f0:66:cf:6d), Dst: c6:ac:47:a3:50:44 (c6:ac:47:a3:50:44)

Destination: c6:ac:47:a3:50:44 (c6:ac:47:a3:50:44)

Address: c6:ac:47:a3:50:44 (c6:ac:47:a3:50:44)

... .. 0 bit: Locally administered address (this is NOT the factory default)

Source: de:d7:f0:66:cf:6d (de:d7:f0:66:cf:6d)

Address: de:d7:f0:66:cf:6d (de:d7:f0:66:cf:6d)

... .. 0 bit: Locally administered address (this is NOT the factory default)

... .. 0 bit: Individual address (unicast)

Type: IPv4 (0x0000)

Internet Protocol Version 4, Src: 10.10.0.33, Dst: 8.8.8.8

0000 ... = Version: 4

... 0101 = Header Length: 20 bytes (5)

Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)

Total Length: 76

Identification: 0x7581 (30081)

0000 ... = Flags: 0x0

... 0000 0000 0000 = Fragment Offset: 0

Time to Live: 128

Protocol: UDP (17)

Header Checksum: 0xaae5 [validation disabled]

Header Checksum status: Unverified

Source Address: 10.10.0.33

Destination Address: 8.8.8.8

User Datagram Protocol, Src Port: 53464, Dst Port: 53

Domain Name System (query)

0000 c6 ac 47 a3 50 44 de d7 f0 66 cf 6d 00 00 45 00 --GPO...f.m--E-

0010 00 4c 75 81 00 00 00 11 aa e5 00 00 00 21 00 00 --L...-i-

0020 00 00 00 00 35 00 38 92 67 9b 1d 01 00 00 01 --...S-g-----

0030 00 00 00 00 00 00 0a 5e 65 78 75 73 72 75 6c 65 --...n exa...rule

0040 73 0a 6f 66 66 69 63 65 61 78 79 73 04 6c 69 76 --s:office app...liv

0050 65 03 63 6f 6d 00 00 01 00 01 e-com...-

It's important that we're able to narrow down our search to the affected source IP(s) in question.

Wireshark has a field called http.host we can search for which allows you to see any network traffic with an associated domain name.

<https://osqa-ask.wireshark.org/questions/33938/how-can-i-filter-by-website-names/>

Setting http.host as a filter for our traffic we end up getting significantly less results which is good for us as we're looking for those IOCs.

http.host									
No.	Time	Source	Destination	Protocol	Length	Info			
2186	257.453066	10.10.0.33	93.184.220.29	HTTP	294	GET /MFewTzBNfEswSTA3BgUrDgKCGuAB8Q50tcX2Fh0Zt1K2B=85iPI7wEwXoD1QQUT1JUIB1V5uNu5gX2F6X2B=ks7QYXjzKCEA177e19ggmle13j64vdGL0K3D HTTP/1.1			
1713	100.949434	10.10.0.33	194.87.236.205	HTTP	377	GET /jollion/lipster.exe HTTP/1.1			
1702	86.588432	10.10.0.33	194.87.236.205	HTTP	278	GET /favicon.ico HTTP/1.1			
1675	86.394989	10.10.0.33	194.87.236.205	HTTP	387	GET /icons/binary.gif HTTP/1.1			
1674	86.394892	10.10.0.33	194.87.236.205	HTTP	385	GET /icons/back.gif HTTP/1.1			
1666	86.350573	10.10.0.33	194.87.236.205	HTTP	386	GET /icons/blank.gif HTTP/1.1			
1660	86.281819	10.10.0.33	194.87.236.205	HTTP	324	GET /jollion/ HTTP/1.1			
1646	82.258695	10.10.0.33	13.107.5.80	HTTP	709	GET /qsm1.aspx?query=http%3A%2F%2Fhypercustom.top%2Fjollion%2F&maxwidth=32765&rowheight=20§ionHeight=160&FORM=IESS02&market=en-US HTTP/1.1			
1641	81.321582	10.10.0.33	104.80.229.162	HTTP	215	GET /pkips/certs/MicSecSerCA2011_2011-10-18.crt HTTP/1.1			
1611	81.173725	10.10.0.33	93.184.220.29	HTTP	290	GET /MFewTzBNfEswSTA3BgUrDgKCGuAB8TBL0V27RVZ7LBDuomK2FnyB455PUeWQUSZ1Z1HJHwPys%28ghuNoZ70rUETFACEA8U118gImZT9XhrH1JQeIK3D HTTP/1.1			
1220	80.374805	10.10.0.33	93.184.220.29	HTTP	296	GET /MFewTzBNfEswSTA3BgUrDgKCGuAB8TBL0V27RVZ7LBDuomK2FnyB455PUeWQUSZ1Z1HJHwPys%28ghuNoZ70rUETFACEA8U118gImZT9XhrH1JQeIK3D HTTP/1.1			
1184	80.055558	10.10.0.33	13.107.5.80	HTTP	706	GET /qsm1.aspx?query=http%3A%2F%2Fhypercustom.top%2Fjollion&maxwidth=32765&rowheight=20§ionHeight=160&FORM=IESS02&market=en-US HTTP/1.1			
1179	78.267298	10.10.0.33	8.238.24.126	HTTP	336	GET /msdownload/update/v3/static/trusted/en/pinruleset1.cab?fa8db18829d4c6ca HTTP/1.1			
1170	73.969011	10.10.0.33	13.107.5.80	HTTP	705	GET /qsm1.aspx?query=http%3A%2F%2Fhypercustom.top%2Fjollion&maxwidth=32765&rowheight=20§ionHeight=160&FORM=IESS02&market=en-US HTTP/1.1			
1166	73.585780	10.10.0.33	13.107.5.80	HTTP	704	GET /qsm1.aspx?query=http%3A%2F%2Fhypercustom.top%2Fjollion&maxwidth=32765&rowheight=20§ionHeight=160&FORM=IESS02&market=en-US HTTP/1.1			
1162	71.914485	10.10.0.33	13.107.5.80	HTTP	703	GET /qsm1.aspx?query=http%3A%2F%2Fhypercustom.top%2Fjollion&maxwidth=32765&rowheight=20§ionHeight=160&FORM=IESS02&market=en-US HTTP/1.1			
1158	71.753437	10.10.0.33	13.107.5.80	HTTP	702	GET /qsm1.aspx?query=http%3A%2F%2Fhypercustom.top%2Fjollion&maxwidth=32765&rowheight=20§ionHeight=160&FORM=IESS02&market=en-US HTTP/1.1			
1155	71.723328	10.10.0.33	13.107.5.80	HTTP	701	GET /qsm1.aspx?query=http%3A%2F%2Fhypercustom.top%2Fjollion&maxwidth=32765&rowheight=20§ionHeight=160&FORM=IESS02&market=en-US HTTP/1.1			
1149	70.297893	10.10.0.33	13.107.5.80	HTTP	387	GET /qsm1.aspx?query=http%3A%2F%2Fhypercustom.top%2Fjollion&maxwidth=32765&rowheight=20§ionHeight=160&FORM=IESS02&market=en-US HTTP/1.1			
1074	42.168940	10.10.0.33	93.184.220.29	HTTP	290	GET /MFewTzBNfEswSTA3BgUrDgKCGuAB8SAUQYBqZawm1Rh6DohK2F8y8gFV7gQUA95QVbRtLtw8KP16xvD17I90VUCEA0LqoXyo4hxxe7H52Fz9DKA3D HTTP/1.1			
661	18.902116	10.10.0.33	194.87.236.205	HTTP	278	GET /favicon.ico HTTP/1.1			
258	18.369575	10.10.0.33	142.250.179.131	HTTP	293	GET /gts1c3/MFewTzBNfEswSTA3BgUrDgKCGuAB8THLnmK3F9HNL067UdCuLV6wCQHYwQinR32F4X0I7gXNPZzQ4kYU83E1H5cCEBtpDExK2Fw4Y3CgAAADyW3QK3D HTTP/1.1			
233	18.159304	10.10.0.33	239.255.255.250	SSDP	179	M-SEARCH * HTTP/1.1			
232	18.158969	10.10.0.33	239.255.255.250	SSDP	179	M-SEARCH * HTTP/1.1			
198	17.901207	10.10.0.33	142.250.179.131	HTTP	293	GET /gts1c3/MFewTzBNfEswSTA3BgUrDgKCGuAB8THLnmK3F9HNL067UdCuLV6wCQHYwQinR32F4X0I7gXNPZzQ4kYU83E1H5cCEBtpDExK2Fw4Y3CgAAADyW3QK3D HTTP/1.1			
196	17.876054	10.10.0.33	142.250.179.131	HTTP	284	GET /gtsr1/MF4wTDK8EgwRJA3BgUrDgKCGuAB8QkccL4d4Lq6j7bE781XZzEbmfwUAQUSK8r3nEak0ghnS95Z1zV81KtC4CDQIDVFNZazTHPUBUGYK3D HTTP/1.1			
191	17.834838	10.10.0.33	142.250.179.131	HTTP	293	GET /gtsr1/MF4wTDK8EgwRJA3BgUrDgKCGuAB8QkccL4d4Lq6j7bE781XZzEbmfwUAQUSK8r3nEak0ghnS95Z1zV81KtC4CDQIDVFNZazTHPUBUGYK3D HTTP/1.1			
171	17.752560	10.10.0.33	2.22.147.10	HTTP	299	GET /msdownload/update/v3/static/trusted/en/disallowedcertstl.cab?cea32afe9aa2a027 HTTP/1.1			
169	17.752351	10.10.0.33	2.22.147.10	HTTP	299	GET /msdownload/update/v3/static/trusted/en/disallowedcertstl.cab?cea32afe9aa2a027 HTTP/1.1			
116	16.504947	10.10.0.33	194.87.236.205	HTTP	316	GET / HTTP/1.1			
109	15.347904	10.10.0.33	239.255.255.250	SSDP	179	M-SEARCH * HTTP/1.1			
108	15.143466	10.10.0.33	239.255.255.250	SSDP	179	M-SEARCH * HTTP/1.1			
107	15.143132	10.10.0.33	239.255.255.250	SSDP	179	M-SEARCH * HTTP/1.1			
84	12.346013	10.10.0.33	239.255.255.250	SSDP	179	M-SEARCH * HTTP/1.1			
82	12.157374	10.10.0.33	239.255.255.250	SSDP	179	M-SEARCH * HTTP/1.1			
81	12.143162	10.10.0.33	239.255.255.250	SSDP	179	M-SEARCH * HTTP/1.1			

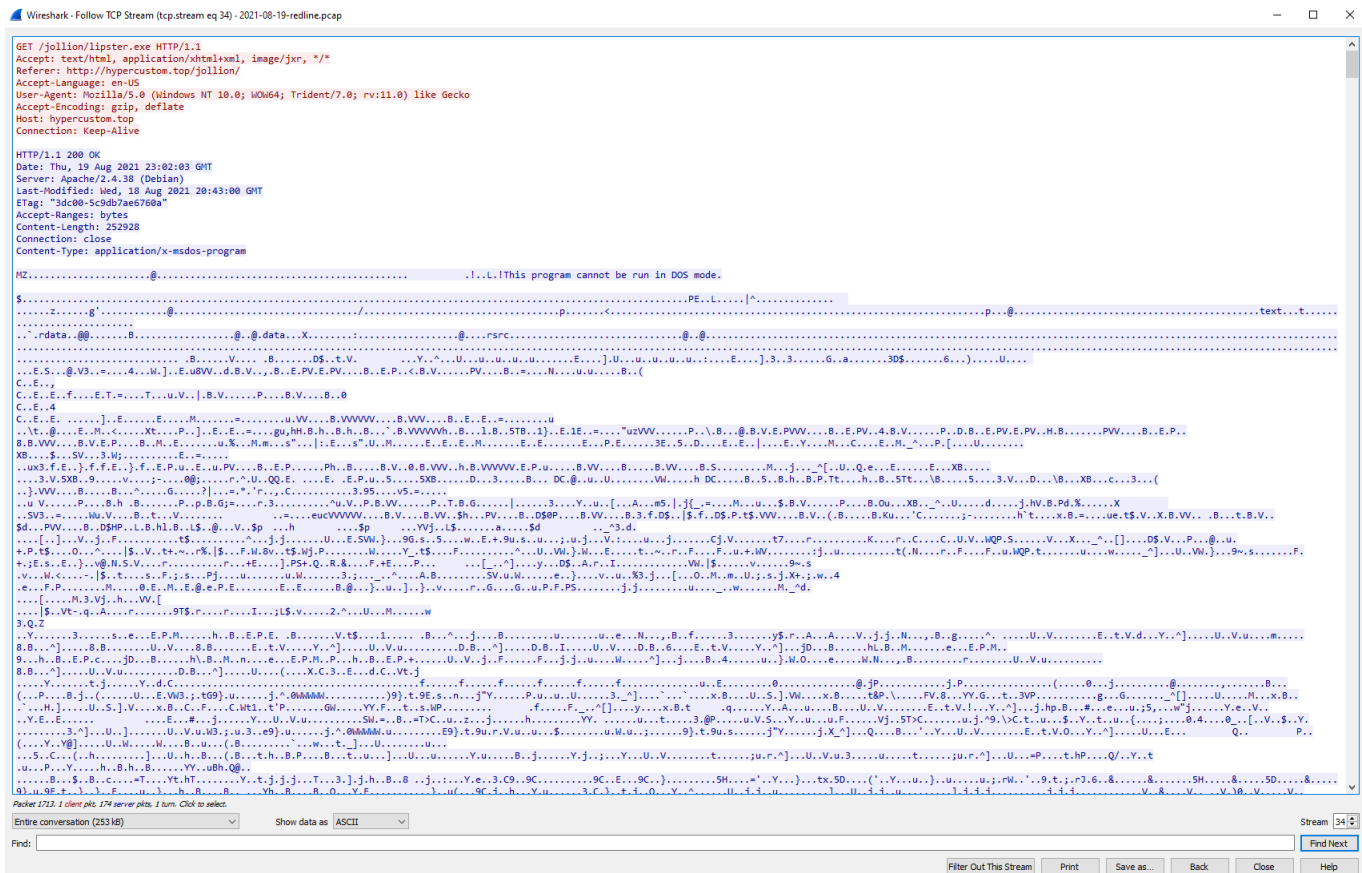
Now, we see only the source IP: 10.10.0.33!

When searching for potentially malicious internet traffic I recommend searching for High Risk IP Addresses: <https://www.fraudlabspro.com/high-risk-ip-address>

Looking over this network traffic I immediately see a GET request of an executable with a destination IP: 194.87.236.205 on Aug 19, 2021 18:02:03.094668000 Central Daylight Time

1713	100.949434	10.10.0.33	194.87.236.205	HTTP	377	GET /jollion/lipster.exe HTTP/1.1
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To gather additional information I right click on the traffic and follow TCP stream bringing us to this window:



Up at the top we see incredibly useful information!

Domain: hypercustom[.]top/jollion/

User-Agent: Mozilla/5.0 (Windows NT 10.0; WOW64; Trident/7.0; rv:11.0) like Gecko

Server: Apache/2.4.38 (Debian)

Executable: /jollion/lipster.exe

OSINT on the domain/executable:

<https://otx.alienvault.com/indicator/domain/hypercustom.top?scrollTo=files>

<https://www.virustotal.com/gui/url/8df3793fa1136d572f7d36399e3ea666eea6c8b4e73d3536d6d994dec6dd76ad/detection>

<https://www.malware.me/analysis/19733/summary/>

This is the malicious domain based off the results from both VT and alienvault.

DOMAIN

hypercustom.top

Add to Pulse

City

REDACTED FOR PRIVACY

Country

DE

Creation Date

2021-08-15T19:41:06

Dissec

unsigned

Domain Name

hypercustom.top

SHOWING 1 TO 10 OF 22 ENTRIES

1 2 3 NEXT

Passive DNS

STATUS	HOSTNAME	QUERY TYPE	ADDRESS	FIRST SEEN	LAST SEEN	ASN	COUNTRY
Unknown	hypercustom.top	A	NOGDMN	2021-08-08 06:16	2024-02-08 02:44		
Unknown	hypercustom.top	A	46.73.214.97	2021-08-24 03:41	2021-08-25 12:06	AS4786 garant-park-internet-fo	Russia
Unknown	hypercustom.top	NS	b.dnspod.com	2021-08-20 12:06	2021-08-20 12:07	ASNone	China
Unknown	hypercustom.top	NS	adnspod.com	2021-08-20 12:06	2021-08-20 12:07	AS134756 chinanet nanjing jidian idc network	China
Unknown	hypercustom.top	SOA	adnspod.com	2021-08-20 12:06	2021-08-20 12:07	AS134756 chinanet nanjing jidian idc network	China
Unknown	hypercustom.top	NS	c.dnspod.com	2021-08-20 12:06	2021-08-20 12:07	AS4837 china unicom china99 backbone	China
Unknown	hypercustom.top	A	194.87.236.205	2021-08-19 11:59	2021-08-20 12:07	AS48347 jsc mediasoft ekspert	Russia

Associated Files

Show 10 entries

DATE	HASH	AVAST	AVG	CLAMAV	MSDEFENDER
Aug 28, 2024	5a8f82f9f6c8803862da7a9d6da6589e70b6f7502050c0438f963a2472cde			WinDropper/Fagtor-9885202-0	
Aug 24, 2021	76089e8324b282280061ba071c0b0a473e8e980a0595a3d3ed9e776507e4	Win32/Malware-gen		WinPacked/Barya-9895531-0	
Aug 23, 2021	ce9877f9b8a40af7b0a476339a0672f87767ec0668f3963a43ba689189950b	Win32/Dropper-gen.(troj)			

SHOWING 1 TO 3 OF 3 ENTRIES

Related Domains

Also, another hint is that the user agent is Internet Explorer. Anything with a user agent of IE needs to be investigated as nothing should be using that program ideally.

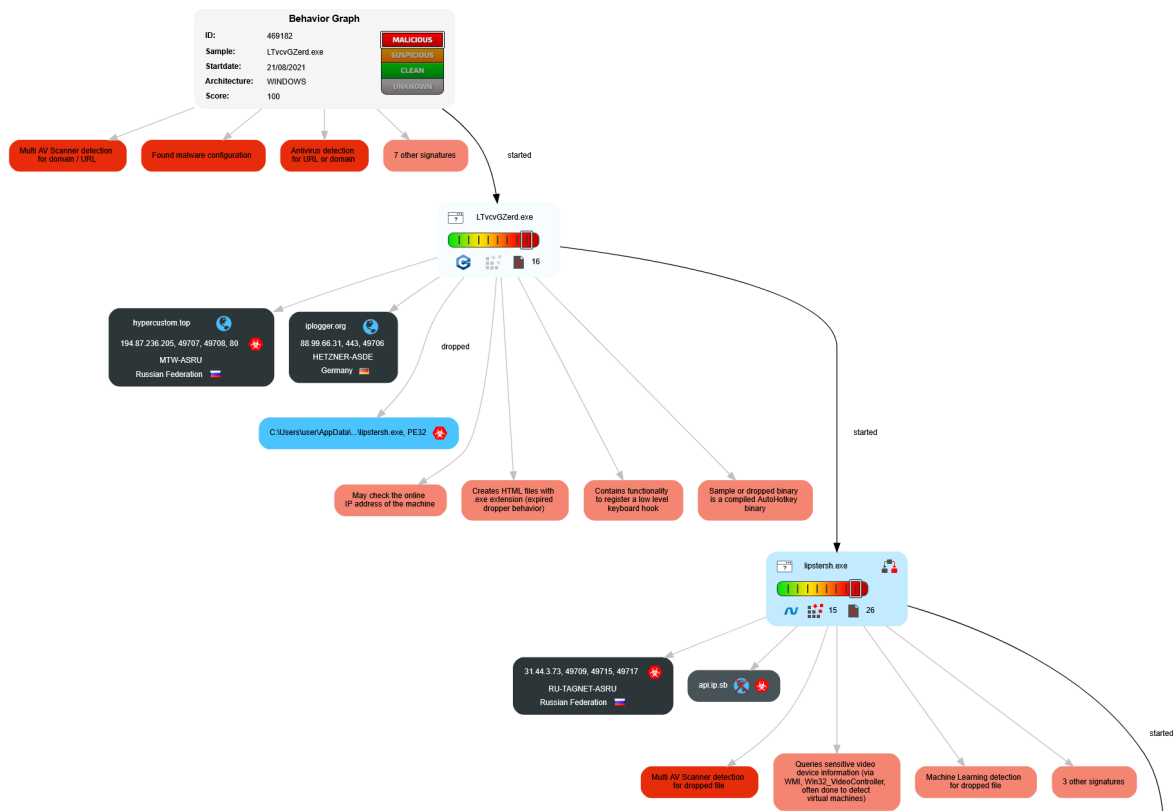
ip.dst == 194.87.236.205

No.	Time	Source	Destination	Protocol	Length	Info
2012	101.149716	10.10.0.33	194.87.236.205	TCP	60	53497 → 80 [ACK] Seq=324 Ack=214501 Win=262144 Len=0
2013	101.149740	10.10.0.33	194.87.236.205	TCP	60	53497 → 80 [ACK] Seq=324 Ack=215961 Win=262144 Len=0
2014	101.149771	10.10.0.33	194.87.236.205	TCP	60	53497 → 80 [ACK] Seq=324 Ack=217421 Win=262144 Len=0
2015	101.149795	10.10.0.33	194.87.236.205	TCP	60	53497 → 80 [ACK] Seq=324 Ack=218881 Win=262144 Len=0
2020	101.179324	10.10.0.33	194.87.236.205	TCP	60	53497 → 80 [ACK] Seq=324 Ack=220341 Win=262144 Len=0
2021	101.179458	10.10.0.33	194.87.236.205	TCP	60	53497 → 80 [ACK] Seq=324 Ack=221801 Win=262144 Len=0
2028	101.180046	10.10.0.33	194.87.236.205	TCP	60	53497 → 80 [ACK] Seq=324 Ack=223261 Win=262144 Len=0
2029	101.180160	10.10.0.33	194.87.236.205	TCP	60	53497 → 80 [ACK] Seq=324 Ack=224721 Win=262144 Len=0
2030	101.180273	10.10.0.33	194.87.236.205	TCP	60	53497 → 80 [ACK] Seq=324 Ack=226181 Win=262144 Len=0
2031	101.180381	10.10.0.33	194.87.236.205	TCP	60	53497 → 80 [ACK] Seq=324 Ack=227641 Win=262144 Len=0
2038	101.181243	10.10.0.33	194.87.236.205	TCP	60	53497 → 80 [ACK] Seq=324 Ack=229101 Win=262144 Len=0
2039	101.181274	10.10.0.33	194.87.236.205	TCP	60	53497 → 80 [ACK] Seq=324 Ack=230561 Win=262144 Len=0
2040	101.181302	10.10.0.33	194.87.236.205	TCP	60	53497 → 80 [ACK] Seq=324 Ack=232021 Win=262144 Len=0
2041	101.181328	10.10.0.33	194.87.236.205	TCP	60	53497 → 80 [ACK] Seq=324 Ack=233481 Win=262144 Len=0
2042	101.181408	10.10.0.33	194.87.236.205	TCP	60	53497 → 80 [ACK] Seq=324 Ack=234941 Win=262144 Len=0
2043	101.181436	10.10.0.33	194.87.236.205	TCP	60	53497 → 80 [ACK] Seq=324 Ack=236401 Win=262144 Len=0
2044	101.181464	10.10.0.33	194.87.236.205	TCP	60	53497 → 80 [ACK] Seq=324 Ack=237861 Win=262144 Len=0
2045	101.181491	10.10.0.33	194.87.236.205	TCP	60	53497 → 80 [ACK] Seq=324 Ack=239321 Win=262144 Len=0
2056	101.181984	10.10.0.33	194.87.236.205	TCP	60	53497 → 80 [ACK] Seq=324 Ack=240781 Win=262144 Len=0
2057	101.182255	10.10.0.33	194.87.236.205	TCP	60	53497 → 80 [ACK] Seq=324 Ack=242241 Win=262144 Len=0
2058	101.182362	10.10.0.33	194.87.236.205	TCP	60	53497 → 80 [ACK] Seq=324 Ack=243701 Win=262144 Len=0
2059	101.183149	10.10.0.33	194.87.236.205	TCP	60	53497 → 80 [ACK] Seq=324 Ack=245161 Win=262144 Len=0
2060	101.183179	10.10.0.33	194.87.236.205	TCP	60	53497 → 80 [ACK] Seq=324 Ack=246621 Win=262144 Len=0
2061	101.183205	10.10.0.33	194.87.236.205	TCP	60	53497 → 80 [ACK] Seq=324 Ack=248081 Win=262144 Len=0
2062	101.183231	10.10.0.33	194.87.236.205	TCP	60	53497 → 80 [ACK] Seq=324 Ack=249541 Win=262144 Len=0
2063	101.183302	10.10.0.33	194.87.236.205	TCP	60	53497 → 80 [ACK] Seq=324 Ack=251001 Win=262144 Len=0
2064	101.183329	10.10.0.33	194.87.236.205	TCP	60	53497 → 80 [ACK] Seq=324 Ack=252461 Win=262144 Len=0
2065	101.183355	10.10.0.33	194.87.236.205	TCP	60	53497 → 80 [ACK] Seq=324 Ack=253921 Win=261400 Len=0
2066	101.228483	10.10.0.33	194.87.236.205	TCP	60	53497 → 80 [FIN, ACK] Seq=324 Ack=253201 Win=261400 Len=0
2075	116.632770	10.10.0.33	194.87.236.205	TCP	60	53495 → 80 [ACK] Seq=1 Ack=214 Win=261928 Len=0
2082	122.205201	10.10.0.33	194.87.236.205	TCP	60	53495 → 80 [FIN, ACK] Seq=1 Ack=214 Win=261928 Len=0
2083	122.205452	10.10.0.33	194.87.236.205	TCP	60	53495 → 80 [RST, ACK] Seq=2 Ack=214 Win=0 Len=0
2087	122.248192	10.10.0.33	194.87.236.205	TCP	60	53495 → 80 [RST] Seq=2 Win=0 Len=0
2118	130.995282	10.10.0.33	194.87.236.205	TCP	60	53498 → 80 [ACK] Seq=1 Ack=214 Win=261928 Len=0
2160	235.414101	10.10.0.33	194.87.236.205	TCP	60	53498 → 80 [FIN, ACK] Seq=1 Ack=214 Win=261928 Len=0
2161	235.414464	10.10.0.33	194.87.236.205	TCP	60	53498 → 80 [RST, ACK] Seq=2 Ack=214 Win=0 Len=0

Protocol: TCP (6)

Header Checksum: 0x11b9 [validation disabled]

Doing a bit of research we actually see that destination IP is associated with Russia and a known IOC of Redline found below!



Now that we found the initial executable that was downloaded along with the domain we need to look for what happened after this occurred.

This leads us to a known Redline C2 domain 185.215.113.17

2130	152.16.0.81	10.10.0.33	185.215.113.17	TCP	66 53515 -> 18597 [SYN] Seq=185 ACK=2383 Win=0 MSS=0 Len=0
2131	177.090651	10.10.0.33	185.215.113.17	TCP	66 53515 -> 18597 [SYN] Seq=185 ACK=2383 Win=0 MSS=0 Len=0
2132	108.109586	10.10.0.33	185.215.113.17	TCP	66 [TCP Retransmission] [TCP Port numbers reused] 53515 -> 18597 [SYN] Seq=0 Win=0 Len=0 MSS=1460 WS=256 SACK_PERM
2133	108.104400	185.215.113.17	10.10.0.33	ICMP	94 Destination unreachable (Host unreachable)
2134	183.154954	185.215.113.17	10.10.0.33	ICMP	94 Destination unreachable (Host unreachable)
2137	108.116762	10.10.0.33	185.215.113.17	TCP	66 [TCP Retransmission] [TCP Port numbers reused] 53515 -> 18597 [SYN] Seq=0 Win=0 Len=0 MSS=1460 WS=256 SACK_PERM
2138	186.268457	185.215.113.17	10.10.0.33	ICMP	94 Destination unreachable (Host unreachable)
2139	233.240372	10.10.0.33	185.215.113.17	TCP	66 53515 -> 18597 [SYN] Seq=0 Win=0 Len=0 MSS=1460 WS=256 SACK_PERM
2140	205.378254	185.215.113.17	10.10.0.33	ICMP	94 Destination unreachable (Host unreachable)
2143	206.351119	10.10.0.33	185.215.113.17	TCP	66 [TCP Retransmission] [TCP Port numbers reused] 53515 -> 18597 [SYN] Seq=0 Win=0 Len=0 MSS=1460 WS=256 SACK_PERM
2144	208.568260	185.215.113.17	10.10.0.33	ICMP	94 Destination unreachable (Host unreachable)
2147	212.351103	10.10.0.33	185.215.113.17	TCP	66 [TCP Retransmission] [TCP Port numbers reused] 53515 -> 18597 [SYN] Seq=0 Win=0 Len=0 MSS=1460 WS=256 SACK_PERM
2148	212.568430	185.215.113.17	10.10.0.33	ICMP	94 Destination unreachable (Host unreachable)
2151	229.384134	10.10.0.33	185.215.113.17	TCP	66 53517 -> 18597 [SYN] Seq=0 Win=0 Len=0 MSS=1460 WS=256 SACK_PERM
2152	232.195160	10.10.0.33	2.18.185.186	TCP	60 53488 -> 443 [FIN, ACK] Seq=876 Ack=6483 Win=0 Len=0
2155	232.286665	10.10.0.33	2.18.185.186	TCP	60 53488 -> 443 [RST, ACK] Seq=876 Ack=6483 Win=0 Len=0
2156	232.287953	10.10.0.33	2.18.185.186	TCP	60 53488 -> 443 [RST] Seq=876 Win=0 Len=0
2157	232.388127	185.215.113.17	10.10.0.33	ICMP	94 Destination unreachable (Host unreachable)
2158	232.387950	10.10.0.33	185.215.113.17	TCP	66 [TCP Retransmission] [TCP Port numbers reused] 53517 -> 18597 [SYN] Seq=0 Win=0 Len=0 MSS=1460 WS=256 SACK_PERM
2159	235.378104	185.215.113.17	10.10.0.33	ICMP	94 Destination unreachable (Host unreachable)
2160	235.414101	10.10.0.33	194.87.236.205	TCP	60 53488 -> 88 [FIN, ACK] Seq=1 Ack=214 Win=261828 Len=0
2161	235.414466	10.10.0.33	194.87.236.205	TCP	60 53488 -> 88 [FIN, ACK] Seq=1 Ack=214 Win=261828 Len=0
2162	238.413590	10.10.0.33	185.215.113.17	TCP	66 [TCP Retransmission] [TCP Port numbers reused] 53517 -> 18597 [SYN] Seq=0 Win=0 Len=0 MSS=1460 WS=256 SACK_PERM
2165	241.468040	185.215.113.17	10.10.0.33	ICMP	94 Destination unreachable (Host unreachable)
2166	255.413963	10.10.0.33	185.215.113.17	TCP	66 53518 -> 18597 [SYN] Seq=0 Win=0 Len=0 MSS=1460 WS=256 SACK_PERM
2167	257.114284	10.10.0.33	8.8.8.8	DNS	80 Standard query 0x1f0e A self-events.data.microsoft.com
2169	257.128457	10.10.0.33	20.189.173.7	TCP	66 53519 -> 443 [SYN] Seq=0 Win=0 Len=0 MSS=1460 WS=8 SACK_PERM
2171	257.273690	10.10.0.33	20.189.173.7	TCP	60 53519 -> 443 [ACK] Seq=1 Ack=1 Win=262144 Len=0

Address: c6rac47:a3:5b:44 (c6rac47:a3:5b:44)

.... 1. = 16 bit: Locally administered address (this is NOT the factory default)

.... 0 = 16 bit: Individual address (unicast)

Source: deid7:f0:66:cf:6d (deid7:f0:66:cf:6d)

Address: deid7:f0:66:cf:6d (deid7:f0:66:cf:6d)

.... 1. = 16 bit: Locally administered address (this is NOT the factory default)

.... 0 = 16 bit: Individual address (unicast)

Type: IPv4 (0x0000)

Internet Protocol Version 4, Src: 10.10.0.33, Dst: 185.215.113.17

0000 = Version: 4

.... 0101 = Header Length: 20 bytes (5)

> Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)

Total Length: 52

Identification: 0x6ac (18156)

010 = Flags: 0x2, Don't fragment

0 = Reserved bit: Not set

1 = Don't fragment: Set

0 = More fragments: Not set

... 0 0000 0000 0000 = Fragment Offset: 0

Time to Live: 128

Protocol: TCP (6)

Header Checksum: 0x7ec4 (validation disabled)

[Header checksum status: Unverified]

Source Address: 10.10.0.33

Destination Address: 185.215.113.17

Transmission Control Protocol, Src Port: 53515, Dst Port: 18597, Seq: 0, Len: 0

Source Port: 53515

Destination Port: 18597

[Stream index: 36]

[Conversation completeness: Incomplete, SYN_SENT (1)]

[TCP Segment Len: 0]

Sequence Number: 0 (relative sequence number)

Sequence Number (raw): 2835818446

[Next Sequence Number: 1 (relative sequence number)]

Acknowledgment Number: 0

Acknowledgment Number (raw): 0

0000 c6 ac 47 a3 5b 44 de d7 f0 66 cf 6d 08 00 45 00 ... 6 PD ... f m E

0010 00 34 46 ec 40 00 88 05 76 c4 00 0a 00 21 00 00 ... 4F ...

0020 70 11 60 48 a5 a9 07 00 5e 00 00 00 00 02 00 00 ... H ...

0030 fa f0 73 c5 00 00 02 04 05 b4 01 03 03 01 01 00 00 ...

0040 04 02

IOC ID:	66366
IOC:	185.215.113.17:80
IOC Type @:	ip:port
Threat Type @:	botnet_cc
Malware:	RedLine Stealer
Malware alias:	RECORDSTEALER
Confidence Level @:	Confidence level is high (100%)
ASN:	AS51381 ELITETEAM-PEERING-AZ1
Country:	SC
First seen:	2021-05-28 19:46:03 UTC
Last seen:	2023-08-01 17:58:45 UTC
UUID:	55e7d393-bfed-11eb-b17b-42010aa4000a
Reporter @:	abuse_ch
Reward @:	5 credits from ThreatFox
Tags:	RedLineStealer

Malware Samples

The table below documents recent malware samples observed that are associated with this indicator of compromise (IOC).

Recommended Actions based on this pcap analysis:

- Contain the host
- Blocklist the domains
- Remove the initial executable
- Monitor the host or other hosts within the environment for connections to either domain*

Thank you for reading my analysis of this pcap and I hope you learned something!

Resources:

https://malpedia.caad.fkie.fraunhofer.de/details/win.redline_stealer

<https://bazaar.abuse.ch/sample/c1659a495f138b63f88811224783a82f96af6f37e6b3df0a8aa771e946a26a7d/#iocs>

<https://tria.ge/210821-bw3kx2chyn/behavioral1>

<https://www.bleepingcomputer.com/tag/redline/>

<https://securityscorecard.com/research/detailed-analysis-redline-stealer/>

<https://threatfox.abuse.ch/ioc/66366/>

Redline pcap: <https://github.com/brett-fitz/malware-pcap/tree/main/redline/pcap>